Power plant and calculations

Power plant and calculation site basically includes the detailed study of power plant operation and maintenance, its related all calculations and thumb rules. It also involves detailed troubleshooting guides for operation and maintenance of power plant system/equipments like Boiler, fans, compressors, belt conveyors, ash handling system, ESP, steam turbine, cooling tower, heat exchangers, steam ejectors, condensers WTP. etc. Heat rate, efficiency Industrial Materials & Equipment

Friday, 25 March 2022

Power plant equipments standard operating procedures (SOP)

SOP: Standard Operating Procedures

SOP is to provide a harmonized organizational framework for the operation of equipments and system.

Objectives of SOP

SOPs are detail-oriented documents and provide step-by-step instructions as to how employees within an organization must go about completing certain tasks and processes.

SOP of Centrifugal Pumps:

START:

For starting of a centrifugal pump proceed as stated below. However, when a pump is put on remote & auto, its suc nd discharge valves are

implemen

- Ensure there are no work permits on equipment & equipment is in normalized condition.
- Ensure tank level is normal.
- Ensure suction valve is open & suction pressure is normal.
- Ensure field all instruments are in service.
- Ensure proper lubrication for pump bearing
- Ensure cooling water flow & pressure to bearings & mechanical seal (If provided)
- Ensure Discharge valve is closed.

Note: For high discharge head pumps, normally discharge valve is in closed condition and there will be minimum recirculation valve. And low discharge head pumps discharge valve is in open condition

- Ensure that all the protection and interlocks are in service.
- Prime the pump by opening air vent from pump body or discharge vent.
- Give start- command by operating LPB or from DCS.
- Once pump achieves normal speed, open discharge valve slowly.
- Check current, vibration, bearing temperature, any noise etc. and record.

SOP STOP:

- To stop a centrifugal pump follow below mentioned steps.
- Keep integral by pass open & close discharge valve, .
- Issue stop command either from local push button or from DCS.
- Keeps cooling water flow some more time to gland cooling etc.
- If the pump is kept as stand by, don’t close CW line valve
- If the pump is stopped for maintenance, isolate suction valve, discharge integral bypass, minimum re-circulation line valve and balancing leak off line valve.
- Stop cooling water flow to pump gland.

SOP of positive displacement pumps:

Plunger pump, reciprocating pump, screw pump& gear pump are such pumps in this category. MOP, LDO pump are gear pumps, lime dosing pump etc. are of screw pumps.

Pumps like HP/LP dosing pumps are plunger pump which are normally used in power plant.

START:

- Follow the steps below for startup of such pump.
- Ensure there are no work permits on equipment & equipment is in normalized condition.
- Ensure normal tank level.
- All field instruments are in service.
- Ensure proper lubrication for pump bearing.
- Ensure suction & discharge valves are open.

Note: Positive displacement pumps should be always started with discharge valve open

- Ensure that integral pump relief valve & that of on discharge line are not gagged.
- Ensure that all the protection and interlocks are in service.

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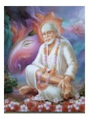

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- Prime the pump by opening air vent from pump body or discharge vent.
- Give start- command by operating LPB or from DCS.
- Check current, vibration, bearing temperature, any noise etc.. and record the data.

SOP STOP:

Switch off keeping all the valves on.

Reference books for power plant O&M

Cooling Water Pump/Tower

To take cooling tower in service, proceed as stated below.

- Ensure that Cooling Tower sump is filled with makeup water up to operating level.
- Tower **distribution** header valves are opened.
- CW return water line valves are open.
- CW pumps are available for operation and free to rotate.
- CT fans are available for operation.
- CT fan gearboxes are filled with requisite lube oil (LO) and up to operating level.
- Protections for CWP and CT fan are in service.
- Ensure field instruments are in service.
- Open suction valve for selected pump, keep discharge valve close.
- Prime pump by opening air release valve on pump.
- Start CW pumps and gradually open the discharge valve.
- Charge cooling water in condenser, **lubricant oil (LO)** cooler, and generator cooler.
- Check discharge pressure, vibration, bearing temperature and motor current.
- Check distribution of cooling water in tower and adjust deck valve to make it uniform.
- Maintain sump level. Put other cooling water pump on auto.

Note:

Before putting any pump on auto, ensure that pump's suction and discharge valves are open and priming of the pump is done.)

SOP Lube Oil cooler & Oil filter change over

SOP Boiler Feed Water Pump start up Fluid Handling

Follow the steps given below

- Ensure there are no work permits on equipment & equipment is in normalized condition.
- Ensure Deaerator level is normal
- Ensure pump suction pressure is normal
- Pump and Motor bearings are properly lubricated and oil cups contain oil.
- Ensure suction strainer differential pressure is normal
- All protection and interlock are in service.
- All field instruments are in service.
- CW line for bearing cooling & gland water cooling is open.
- Open suction valve, keep discharge valve close.
- Ensure minimum re circulation and balancing leak off line valves are open.
- Start pump by operating Local start Push Button or from DCS.
- Ensure pump has reached 50% RPM in just 10 seconds to avoid bearing failure & rubbing of balance & counter balance

discs.

- Gradually open discharge valve. If integral by pass valve is provided, open by pass valve first. Once Feed line is pressurized, main discharge valve can be opened fully.
- Check all parameters, bearing temperature, vibrations, suction / discharge pressure, balance leak off pressure, motor winding temperature etc.
- Put other pump(s) on auto.

Read power plant O&M reference books

SOP Instrument Air Compressor (Reciprocating)

Follow the steps given bellow to start a compressor

- Ensure there are no work permits on equipment & equipment is in normalized condition.
- All protection and interlock are in service.
- All field instruments are in service.
- Sufficient lubricant oil (LO) is present in crank sump.
- CW is charged in compressor and after cooler and flow switch is reset.
- Moisture Traps are in service.
- Suction air filter is cleaned.
- Discharge valve is open.
- Start compressor.
- Check loading and unloading of compressor.
- Record loading unloading current and pressure.
- Check discharge temperature.

Pump Change Over SOP:

- Ensure that stand by pump is available for operation.
- Prime the pump after opening its suction valve. (Where minimum re circulation line is provided, open re circulation line valve).
- Start pump either from local or from remote.
- Open discharge valve and close minimum re circulation line valve.
- Gradually close discharge valve of previous pump. (where minimum re circulation line is provided , open re circulation line valve before closing discharge valve)
- Ensure that incoming pump is maintaining all parameters like discharge pressure, bearing temperature & vibrations.
- Switch off the previous pump.

Boiler Gauge Glass Operation

Initial line up SOP

- Crack open the steam cock. Wait for some time till the gauge glass is heated slowly.
- Crack open the water cock.
- Close the drain cock & vent valve.
- Open the steam cock and water cock fully.

SOP for Gauge glass flushing

- Close the water cock and open the drain cock for some time.
- Close the drain cock and open the water cock.

Note: Water should return to its normal working level quickly. If this does not happen, then there is a blockage in the waterside

- Then Close the steam cock and open the drain cock for some time.
- Close the drain cock and open the steam cock.

Fluid Handling

Note: If the water does not return to its normal working level quickly, then there is blockage in the steam side.

Charging of Main Steam Line

This procedure is applicable when main steam line is depressurized and first boiler is coming into service. Proceed as follows for charging main steam line.

- Ensure that boiler has achieved at least 60% of its rated pressure
- (**for detail operations of boiler refer boiler operation from OEM manual**).
- Ensure that that all the drain line valves are open along with stream trap by pass.
- Open MSSV by pass partially so that line steam pressure is at 5 kg/ cm2.
- Heat up line at least for 10 min.
- Increase pressure up to 10 kg/ cm2 for another 10 min. To control main steam pressure line, warm-up vent can be opened and regulated.
- Once dry steam is observed coming out from all the drains, close the steam trap by pass and keep all the traps in service.
- Increase pressure of the steam line near about the boiler pressure and open MSSV. **Before operating MSSV, care to taken that steam pressure before and after MSSV is more or less same. To match the steam pressure warm up vent can be throttled.**
- Increase pressure and temperature as required for turbine start up.
- It is to be noted that increase of temperature up to 300 deg C should be 3-5 deg C per minute. Above 300 degC the increase of main steam line should be 5-7 degC per minute.

Air Compressors

SOP Ejector System

Normally ejector system consists of one set of Hogger and 2 main ejectors. For ACC ejector system there is

One set of Hogger system and 2 set of service ejector. Each service ejector set consists of 2 nos of ejectors, 1 in 1st stage and 1no ejector in 2nd stage.

To take Hogger ejector in service follow following guidelines

- Ensure auxiliary steam is available at desired pressure and temperature

- Ensure vacuum breaker valve fitted on steam condenser is closed
- Ensure cooling water is circulating in the condenser and the turbine gland is charged fully at 0.1 kg/cm2
- Note: During cold start up gland sealing is charged after taking hogger into line (after vacuum pulling) i.e say at vacuum -0.2 to -0.5 kg/cm2 & During hot start up gland sealing is charged before taking hogger into line (before vacuum pulling).
- Ensure live steam line to ejector steam lines drain are kept open.
- Charge the main steam line to ejector steam & temperature control valve.
- Ensure the rated pressure (10 kg/cm2 or as specified by OEM) and temperature.
- Once the rated parameters are reached open the steam valve of starting or hogger ejector.
- Observe the steam is vented to the atmosphere.
- Then open the ejector airline valve.
- Observe vacuum inside the condenser increasing slowly and will reach 60 to 70% of rated vacuum within 20 minutes

Read More SOPs.

Emergency operation guide in power plants

Slop fired Boiler SOP

Turbine oil flushing SOP

Coal handling plant SOP

Boiler refractory dry out SOP

Boiler feed pump start/stop SOP

Steam turbine cold, warm & hot start up SOPS

Boiler Alkali Boil out SOPs

Boiler Hydraulic test SOP

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